

SHAP, United Kingdom

WMO: 032250

Lat: 54.5022N

Lon: 2.6850W

Elev: 837

StdP: 14.26

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.8	23.5	18.2	14.2	21.7	21.5	16.7	26.4	34.7	45.3	31.8	45.1	1.4	220	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	73.4	63.0	69.8	61.1	66.6	59.5	64.3	71.1	62.5	67.9	60.8	65.0	8.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	85.3	67.0	60.2	80.4	64.9	58.7	76.0	62.9	29.8	71.1	28.5	68.0	27.3	65.1	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	28.9	25.1	21.9	DB	14.7	78.0	4.7	3.3	11.3	80.4	8.5	82.3	5.9	84.1	2.5	86.5
(5)				WB	14.6	67.7	4.4	2.0	11.4	69.2	8.8	70.3	6.3	71.4	3.1	72.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	37.5	37.9	39.9	43.6	48.5	53.5	57.2	56.5	52.3	47.6	42.0	38.2
	DBStd	8.83	5.88	5.73	5.42	5.05	5.48	4.49	3.97	4.13	4.33	5.52	5.97	6.98
	HDD50	2091	386	340	314	198	94	15	1	2	23	109	244	365
	HDD65	6837	851	760	779	641	513	346	244	264	381	538	690	830
	CDD50	735	0	0	1	7	46	119	224	205	92	36	4	1
	CDD65	4	0	0	0	0	0	2	2	2	0	0	0	0
	CDH74	75	0	0	0	0	2	11	35	27	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	9.0	10.7	10.8	10.0	8.7	8.6	7.9	7.2	7.4	7.9	9.2	9.3	10.3
Precipitation	PrecAvg	55.9	6.7	5.1	3.9	3.2	3.3	3.2	3.7	4.1	4.1	5.5	6.1	7.1
	PrecMax	73.0	10.6	13.4	8.5	6.1	5.9	7.4	8.9	10.4	7.7	10.4	16.3	20.2
	PrecMin	40.4	1.1	0.9	1.3	1.2	0.8	1.2	2.0	0.7	0.8	1.8	2.5	1.7
	PrecStd	8.5	2.8	3.3	1.5	1.1	1.6	1.6	1.5	2.0	1.5	2.3	2.9	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	52.4	57.8	66.3	72.2	75.2	77.8	77.5	69.3	62.5	56.5	52.8
		MCWB	48.7	48.2	49.4	53.8	61.6	63.3	65.2	64.7	62.1	56.7	53.9	51.1
	2%	DB	49.3	49.4	53.0	59.4	67.3	69.9	73.9	71.6	65.0	59.3	54.1	51.2
		MCWB	47.3	46.8	48.0	51.4	58.1	60.9	63.2	63.1	59.2	56.4	52.2	49.5
	5%	DB	47.4	47.6	50.2	55.4	62.6	66.2	70.2	67.2	62.2	57.6	52.4	49.5
		MCWB	45.4	45.3	46.2	49.4	55.1	58.6	61.5	60.7	57.3	55.1	50.5	47.8
	10%	DB	45.7	46.0	48.1	52.6	58.5	62.7	66.5	64.4	60.1	55.9	50.7	47.6
		MCWB	43.8	43.9	45.0	47.7	52.5	56.3	60.0	59.1	55.9	53.5	48.8	45.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	49.3	51.5	55.5	62.8	65.6	67.0	66.2	63.6	59.2	54.6	51.7
		MCDB	50.4	51.0	55.9	64.1	70.7	72.6	75.0	73.7	68.0	60.6	55.7	52.4
	2%	WB	47.6	47.5	48.8	52.5	58.8	61.9	64.5	63.7	60.6	57.4	52.7	50.0
		MCDB	48.9	49.1	51.9	57.6	65.9	68.4	71.8	69.7	63.4	58.6	53.8	51.2
	5%	WB	45.9	45.8	47.1	50.3	56.0	59.2	62.5	62.0	58.5	55.6	50.8	48.1
		MCDB	47.2	47.3	49.5	54.4	61.2	64.7	68.3	66.2	61.1	57.2	52.2	49.4
	10%	WB	44.0	44.3	45.5	48.5	53.6	57.3	60.6	60.2	56.7	53.9	49.0	46.2
		MCDB	45.4	45.9	47.8	52.1	57.8	61.7	65.3	63.4	59.4	55.7	50.5	47.7
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.8	12.1	14.7	15.8	15.0	15.1	14.3	13.9	11.8	10.1	9.8
		MCDBR	9.1	11.0	15.7	20.3	23.4	22.2	23.6	20.3	17.1	12.3	10.4	9.8
		MCWBR	9.3	9.4	11.6	13.9	15.4	14.4	14.3	12.9	11.7	10.2	9.8	9.7
	5% WB	MCDBR	9.1	9.6	13.3	18.0	20.5	19.7	19.7	16.8	13.8	10.4	9.9	9.6
		MCWBR	9.5	9.1	10.7	12.9	14.0	13.5	12.8	11.4	10.8	9.5	9.7	10.0
Clear-Sky Solar Irradiance	taub	0.314	0.326	0.353	0.373	0.377	0.375	0.376	0.364	0.349	0.334	0.310	0.298	
	taud	2.285	2.369	2.316	2.286	2.319	2.348	2.348	2.394	2.442	2.441	2.360	2.260	
	Ebn at Noon	197	241	259	270	275	277	274	271	260	236	199	175	
	Edh at Noon	18	24	31	36	37	37	36	33	27	22	17	15	
All-Sky Solar Radiation	RadAvg	175	389	744	1186	1541	1557	1505	1211	857	457	223	134	
	RadStd	23	44	68	116	119	154	156	89	75	56	23	14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page